

Monday - 02-03-2025

Topic: Review of thermodynamics & gas dynamics
remember energy Eqⁿ:

$$-w + \cancel{q} + c_p T_1 + \frac{u_1^2}{2} = c_p T_2 + \frac{u_2^2}{2}$$

for isentropic flow,

$w=0$ for without work $\Rightarrow$

$$\left(c_p T_1 + \frac{u_1^2}{2} = c_p T_2 + \frac{u_2^2}{2} \right)$$

$$c_p T_{o1} = c_p T_{o2}$$

$\Rightarrow T_{o1} = T_{o2}$ & from isentropic relation:

$$\frac{P_{o2}}{P_{o1}} = \left(\frac{T_{o2}}{T_{o1}} \right)^{\frac{\gamma}{\gamma-1}} \Rightarrow P_{o1} = P_{o2}$$

So for isentropic flow without work, $P_{o1} = P_{o2}$ & $T_{o1} = T_{o2}$

This models the rocket nozzles!

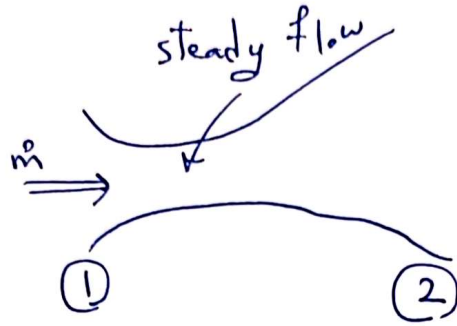
Isentropic flow with work:

$$\text{Energy Eq}^n: c_p T_{o1} - w = c_p T_{o2} \Rightarrow T_{o1} \neq T_{o2} \Rightarrow P_{o1} \neq P_{o2}$$

This models the turbo-machinery of rockets in liquid rockets.

"Isentropic flow without work", $q=0, w=0$

$$\begin{cases} T_{01} = T_{02} = T_0 \\ P_{01} = P_{02} = P_0 \end{cases}$$



The $\dot{m}$ in the rocket nozzle is given by:

$$\dot{m} = \sqrt{\frac{\gamma}{R}} \frac{P_0}{\sqrt{T_0}} M \left(1 + \frac{\gamma-1}{2} M^2\right)^{\frac{-(\gamma+1)}{2(\gamma-1)}} A \quad (E.1) \quad R = \frac{\bar{R}}{M}$$

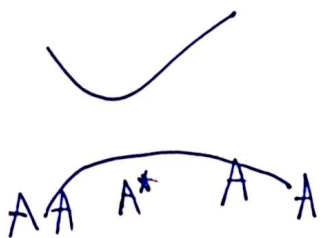
Pay attention to gas species (γ & R depend on species
 $R = \frac{\bar{R}}{M}$)

The flow where $M=1$ is known as critical state, the properties are said to be critical properties & are designated with a superscript (*), i.e., P^*, T^*, A^*, a^*

Evaluating E.1 @ sonic state ($M=1, A=A^*$)

$$\dot{m} = \sqrt{\frac{\gamma}{R}} \frac{P_0}{\sqrt{T_0}} \left(\frac{\gamma+1}{2}\right)^{\frac{-(\gamma+1)}{2(\gamma-1)}} A^* \quad E.2 \quad \text{we can find } A^* \text{ from E.2}$$

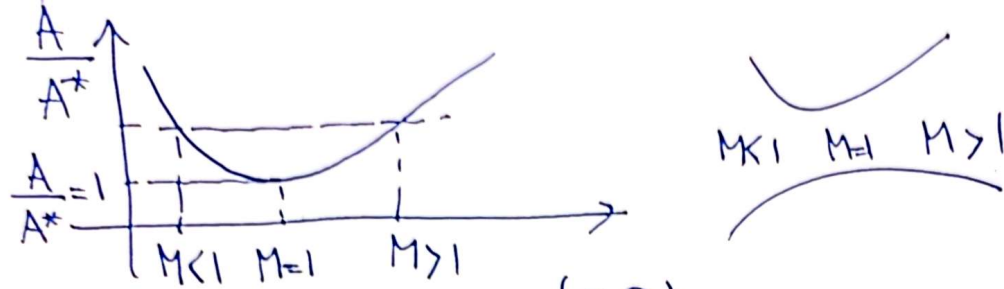
$$A^* = \dot{m} \sqrt{\frac{R}{\gamma}} \frac{\sqrt{T_0}}{P_0} \left(\frac{\gamma+1}{2}\right)^{\frac{\gamma+1}{2(\gamma-1)}} \quad E.3 \quad \text{gives } A^*$$



P-2

$\dot{m} = \text{cte} \rightarrow$ we can omit $\dot{m}$ from E-1 & E.2

$$\frac{E.1}{E.2} \Rightarrow \frac{A}{A^*} = \frac{1}{M} \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M^2\right) \right]^{\frac{\gamma+1}{2(\gamma-1)}}$$



A convergent-divergent ^(C-D) nozzle accelerates the flow from $M < 1$ to $M > 1$. Mach at throat is usually 1.

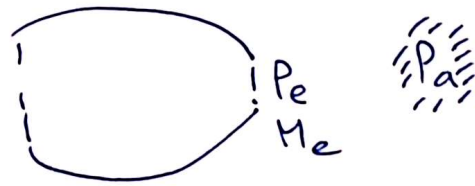
A convergent nozzle accelerates the flow from $M < 1$ to $M = 1$.

$M = 1 \rightarrow$ the flow is choked.

Convergent nozzle

if $M_e < 1 \rightarrow P_e = P_a$

if $M_e = 1 \rightarrow P_e > P_a$



Note for subsonic outlet, $P_e = P_a$ always

C-D nozzle

if $M_e < 1 \rightarrow P_e = P_a$

if $M_e > 1 \rightarrow P_e$ may not be equal P_a

$P_e < P_a$ (over-expanded jet)

launch-time

$P_e = P_a$ (fully - " ")

intermediate flight

$P_e > P_a$ (under - " ")

high altitude

(P-31)

"Isentropic flow with work" ($q=0$, $w \neq 0$)

$w > 0$ for turbine, $w < 0$ for compressor & pump.

Note: ideal gas law should not be used for pumps! as the pump uses liquids, & the fluid in pumps is incompressible.

An adiabatic & reversible pump is called "ideal pump", which is isentropic.

ideal pump:

Incompressible ($\rho = \text{cte}$)

remember from thermodynamics: $T ds = dh - \frac{dp}{\rho}$ Stagnation Point $\rightarrow$

$\underbrace{T ds}_{\text{isentropic}} = dh_o - \frac{dp_o}{\rho} \Rightarrow dh_o = \frac{dp_o}{\rho} \Rightarrow \Delta h_o = \frac{\Delta p_o}{\rho}$ (1)

1st law of thermodynamics: $h_{o2} - h_{o1} = \underbrace{q}_{\text{isentropic}} - w_p \Rightarrow \Delta h_o = -w_p$ (2)

From (1) & (2) $\Rightarrow w_p = \frac{-\Delta p_o}{\rho}$ ideal pump work